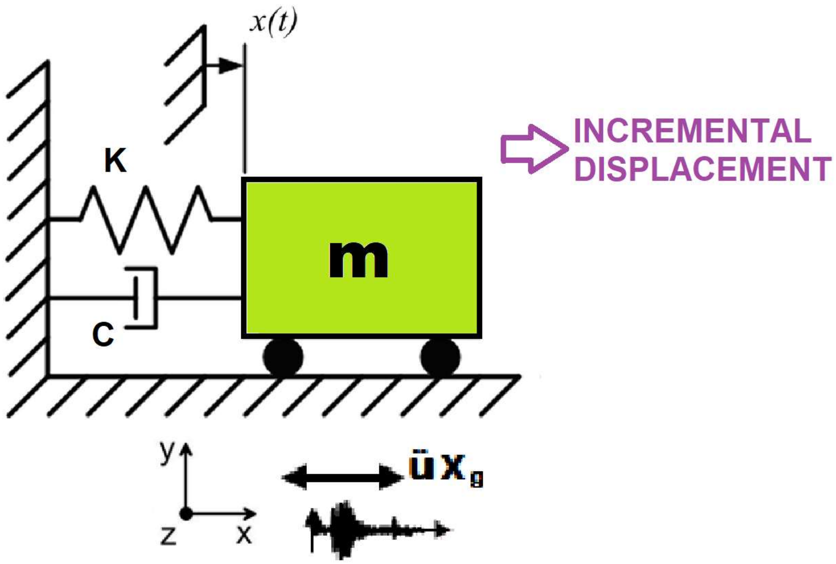
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

# DETERMINATION OF OPTIMUM STRUCTURAL YIELD STRENGTH FROM ENERGY DISSIPATION CAPACITY INDEX USING FINITE-DIFFERENCE NEWTON ITERATION AND OPENSEES

THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)




$$Structural\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

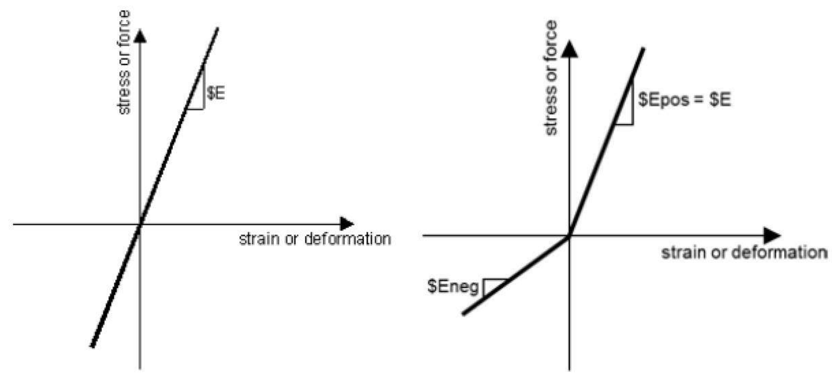
$$\Delta_d = Lateral\ Displaement\ from\ Dynamic\ Analysis$$

$$\Delta_y = Lateral\ Yield\ Displaement\ from\ Pushover\ Analysis$$

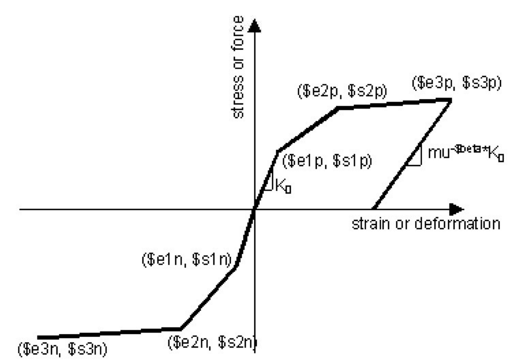
$$\Delta_u = Lateral\ Ultimate\ Displaement\ from\ Pushover\ Analysis$$

$$P(t) = P_0 e^{-0.05\bar{\omega}t} sin(\bar{\omega}t)$$

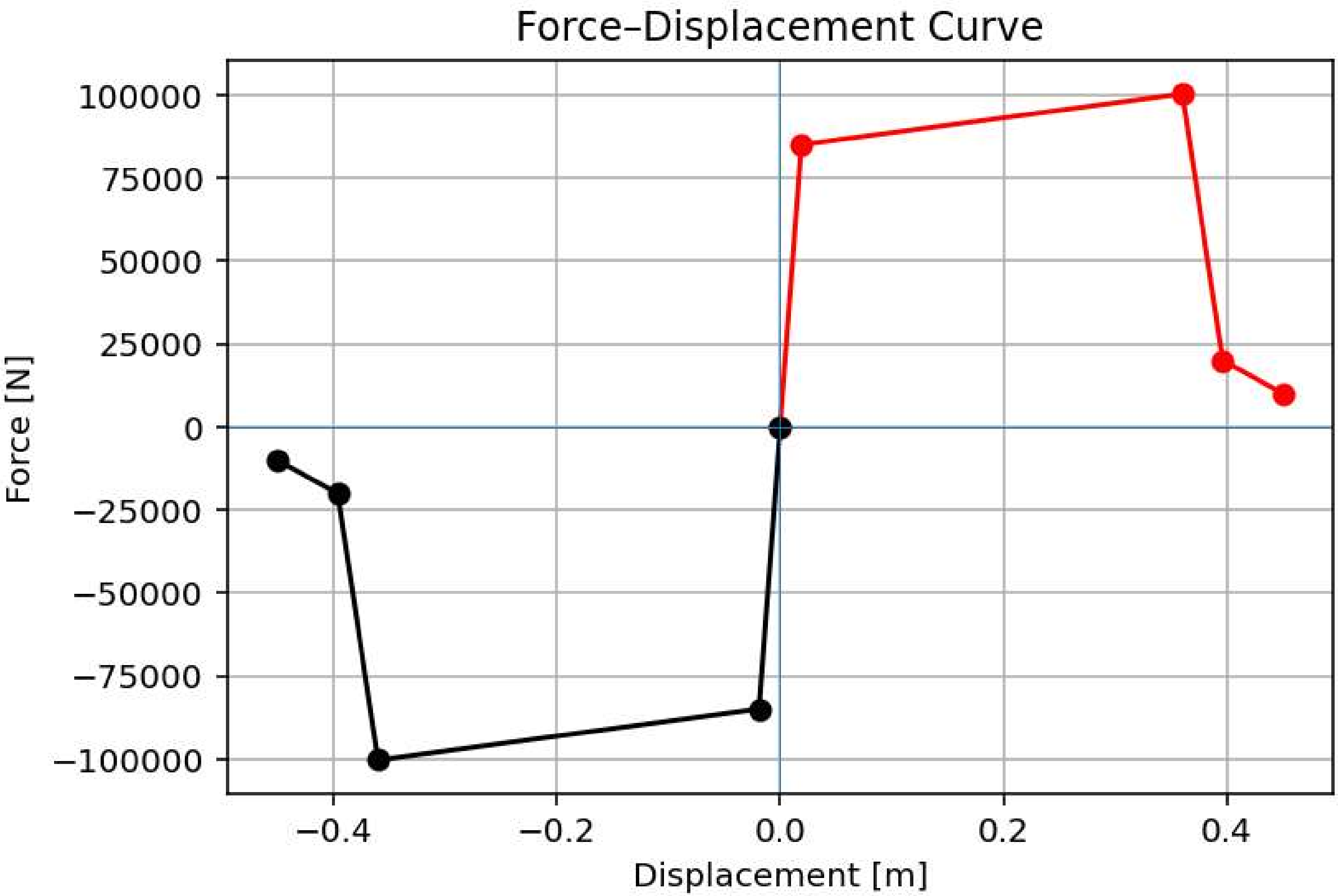
$$m\ddot{u} + c\dot{u} + ku = P(t)$$

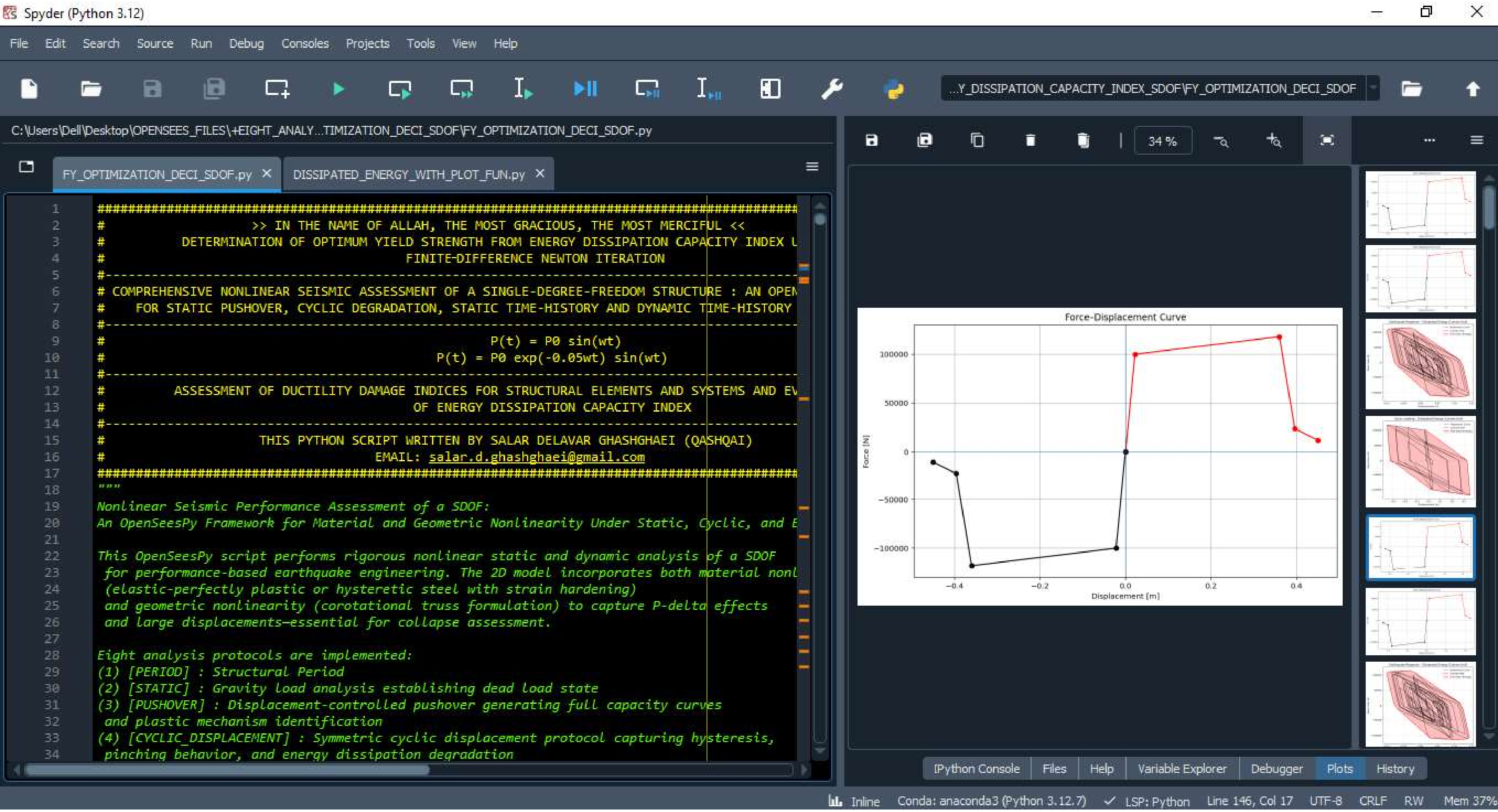



**ELASTIC MATERIAL**




**INELASTIC MATERIAL**


Force–Displacement Curve
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
Force [N]
−0.4
−0.2
0.0
0.2
0.4
Displacement [m]


Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
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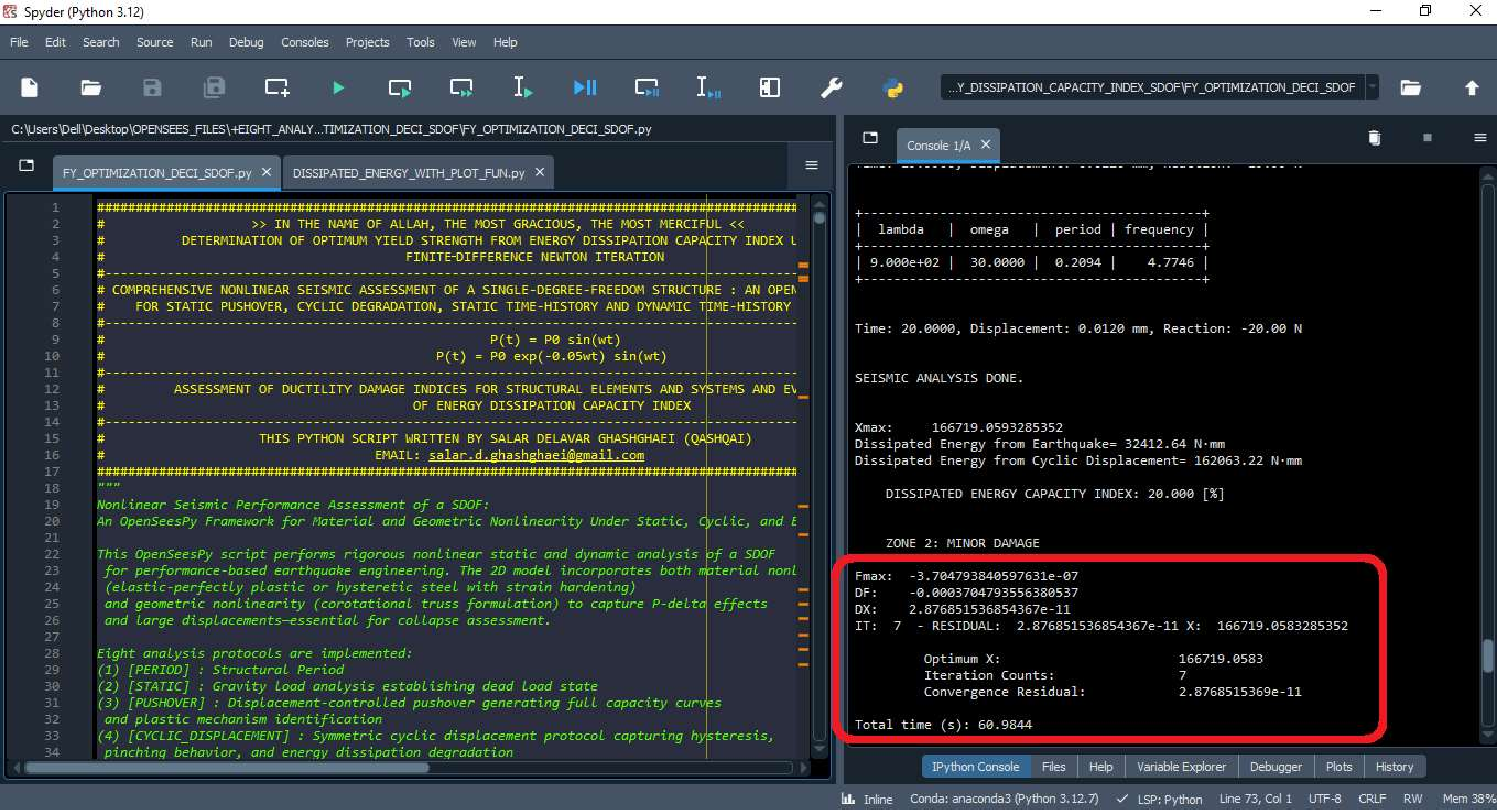
FY_OPTIMIZATION_DECI_SDOF.py
DISSIPATED_ENERGY_WITH_PLOT_FUN.py
# >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
# DETERMINATION OF OPTIMUM YIELD STRENGTH FROM ENERGY DISSIPATION CAPACITY INDEX U
# FINITE-DIFFERENCE NEWTON ITERATION
# COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A SINGLE-DEGREE-FREEDOM STRUCTURE : AN OPEN
# FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY
# P(t) = P0 sin(wt)
# P(t) = P0 exp(-0.05wt) sin(wt)
# ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AND SYSTEMS AND EV
# OF ENERGY DISSIPATION CAPACITY INDEX
# THIS PYTHON SCRIPT WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
# EMAIL: salar.d.ghashghaei@gmail.com
Nonlinear Seismic Performance Assessment of a SDOF:
An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Static, Cyclic, and E
This OpenSeesPy script performs rigorous nonlinear static and dynamic analysis of a SDOF
for performance-based earthquake engineering. The 2D model incorporates both material nonl
(elastic-perfectly plastic or hysteretic steel with strain hardening)
and geometric nonlinearity (corotational truss formulation) to capture P-delta effects
and large displacements—essential for collapse assessment.
Eight analysis protocols are implemented:
(1) [PERIOD] : Structural Period
(2) [STATIC] : Gravity load analysis establishing dead load state
(3) [PUSHOVER] : Displacement-controlled pushover generating full capacity curves
and plastic mechanism identification
(4) [CYCLIC_DISPLACEMENT] : Symmetric cyclic displacement protocol capturing hysteresis,
pinching behavior, and energy dissipation degradation
34 %
Force-Displacement Curve
Force [N]
Displacement [m]
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 146, Col 17 UTF-8 CRLF RW Mem 37%

# OPTIMIZATION PROBLEM FOR ENERGY DISSIPATION CAPACITY INDEX (EDCI)


Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
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C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...TIMIZATION_DECI_SDOF\FY_OPTIMIZATION_DECI_SDOF.py
FY_OPTIMIZATION_DECI_SDOF.py
DISSIPATED_ENERGY_WITH_PLOT_FUN.py
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"""
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Console 1/A
| lambda | omega | period | frequency |
| 9.000e+02 | 30.0000 | 0.2094 | 4.7746 |
Time: 20.0000, Displacement: 0.0120 mm, Reaction: -20.00 N
SEISMIC ANALYSIS DONE.
Xmax: 166719.0593285352
Dissipated Energy from Earthquake= 32412.64 N·mm
Dissipated Energy from Cyclic Displacement= 162063.22 N·mm
DISSIPATED ENERGY CAPACITY INDEX: 20.000 [%]
ZONE 2: MINOR DAMAGE
Fmax: -3.704793840597631e-07
DF: -0.0003704793556380537
DX: 2.876851536854367e-11
IT: 7 - RESIDUAL: 2.876851536854367e-11 X: 166719.0583285352
Optimum X: 166719.0583
Iteration Counts: 7
Convergence Residual: 2.8768515369e-11
Total time (s): 60.9844
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 73, Col 1 UTF-8 CRLF RW Mem 38%

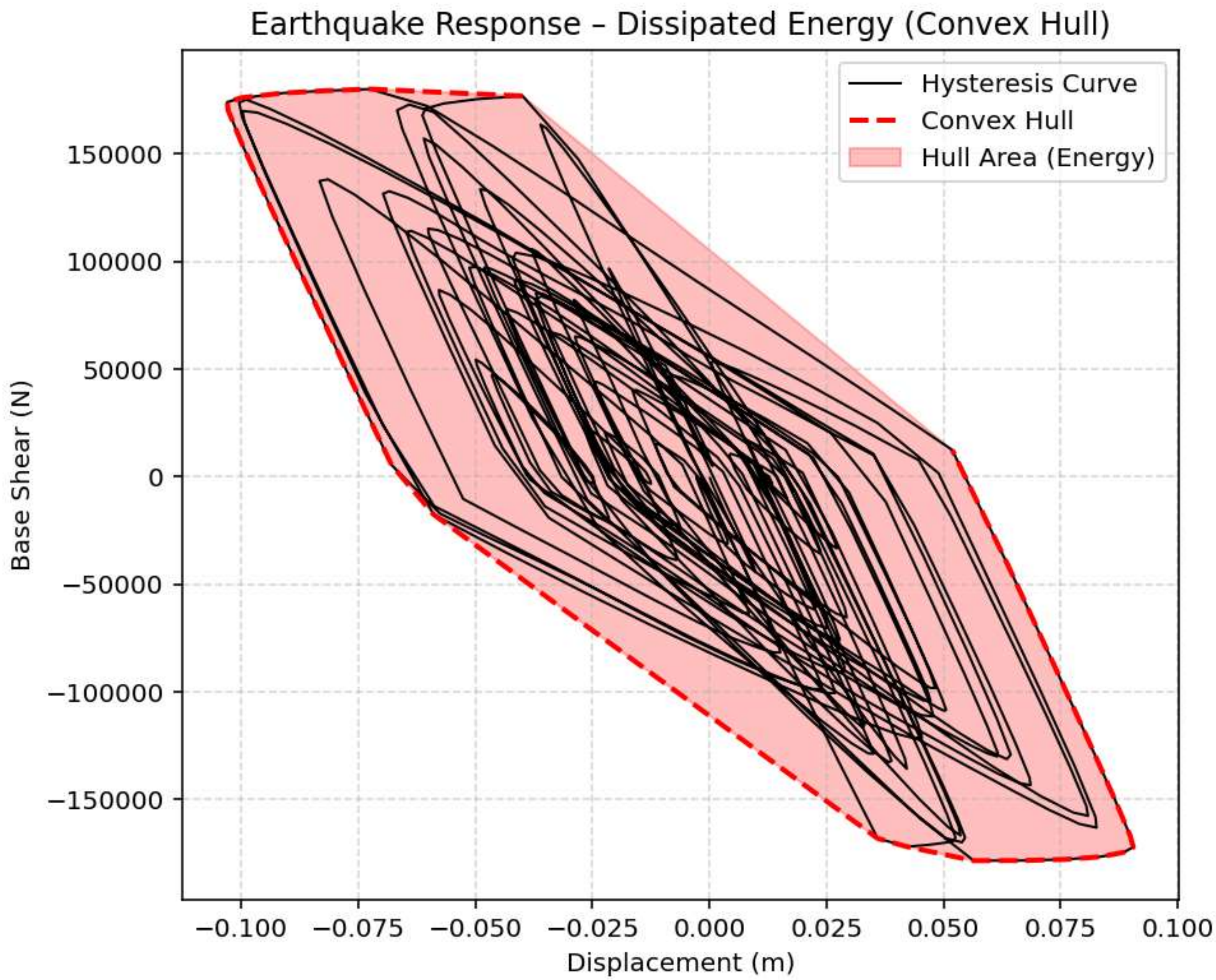
**Dissipated Energy from Earthquake= 32412.64 N·m**
**Dissipated Energy from Cyclic Displacement= 162063.22 N·m**

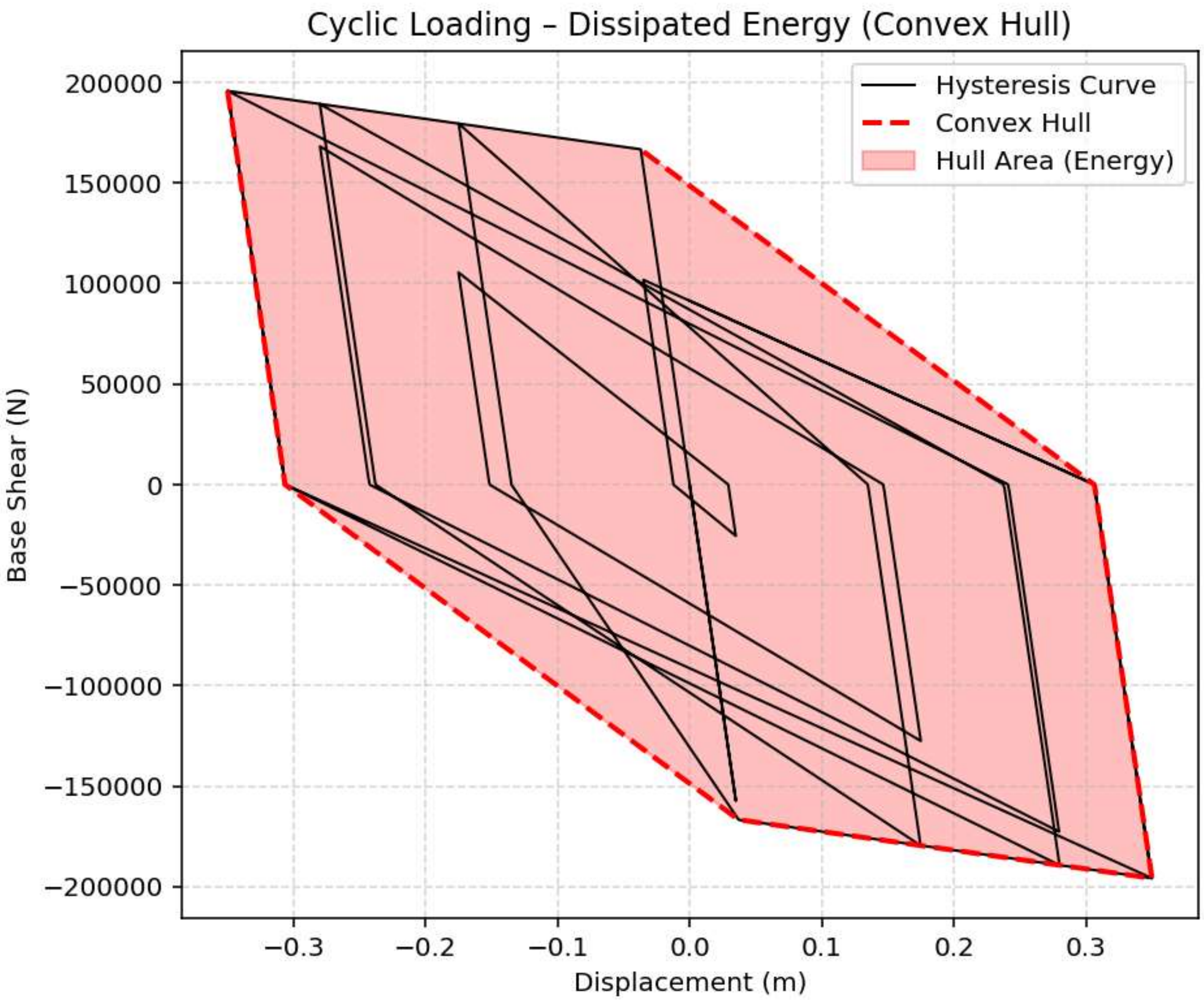
**DISSIPATED ENERGY CAPACITY INDEX: 20.000 [%]**

**ZONE 2: MINOR DAMAGE**

**Fmax: -3.704793840597631e-07**
**DF: -0.0003704793556380537**
**DX: 2.876851536854367e-11**
**IT: 7 - RESIDUAL: 2.876851536854367e-11 X: 166719.0583285352**

| | |
|---|---|
| **Optimum X:** | **166719.0583** |
| **Iteration Counts:** | **7** |
| **Convergence Residual:** | **2.8768515369e-11** |


Earthquake Response – Dissipated Energy (Convex Hull)
Hysteresis Curve
Convex Hull
Hull Area (Energy)
Base Shear (N)
150000
100000
50000
0
−50000
−100000
−150000
−0.100
−0.075
−0.050
−0.025
0.000
0.025
0.050
0.075
0.100
Displacement (m)


Cyclic Loading – Dissipated Energy (Convex Hull)
Hysteresis Curve
Convex Hull
Hull Area (Energy)
Base Shear (N)
200000
150000
100000
50000
0
−50000
−100000
−150000
−200000
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Displacement (m)